

Gradient Descent

Collection of essential background on gradient descent

Following [1], for $f : \mathbb{R}^p \rightarrow \mathbb{R}$ we consider the optimisation problem

$$\min_{x \in \mathbb{R}^p} f(x) \quad (1)$$

under certain assumptions on the function f .

The basic intuition for why an algorithm based on gradient steps can work is the following. Let $f \in C^1(\mathbb{R}^p, \mathbb{R})$. By Taylor-Lagrange, we have

$$\begin{aligned} \forall x, y \in \mathbb{R}^p : f(y) &= f(x) - \langle \nabla f(x), y - x \rangle + o(\|y - x\|) \\ \therefore y = x - \alpha \nabla f(x) &\Rightarrow f(x - \alpha \nabla f(x)) = f(x) - \alpha \|\nabla f(x)\|^2 + o(\alpha) \end{aligned}$$

hence for $\alpha > 0$ small enough, $f(x - \alpha \nabla f(x)) < f(x)$.

Preliminaries

Definition 1. (Convex functions) A function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is **convex** if $\text{dom } f \subseteq \mathbb{R}^p$ is a convex set and

$$\forall x, y \in \mathbb{R}^p, \forall t \in [0, 1] : f((1-t)x + ty) \leq (1-t)f(x) + tf(y). \quad (2)$$

A function is μ -**strongly convex** if $f(x) - \frac{\mu}{2}\|x\|^2$ is a convex function, that is

$$\forall x, y \in \mathbb{R}^p, \forall t \in [0, 1] : f((1-t)x + ty) \leq (1-t)f(x) + tf(y) - \frac{\mu t(1-t)}{2} \|x - y\|^2.$$

Used less often is the notation of **strict** convexity, where the convexity inequality holds strictly.

Lemma 2. (All convex functions are continuous) Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be convex. The $f \in C(\mathbb{R}^p, \mathbb{R})$.

Proof. Via a ‘bowtie’ around a point and then using the squeeze law. ■

Lemma 3. (C^1 convex functions and minima)

If $f \in C^1(\mathbb{R}^p, \mathbb{R})$ is convex, then

1. f convex $\Rightarrow$ local minima are global minima ($\nabla f(x^*) = 0$ in [Lemma 4](#))
2. f strictly convex $\Rightarrow$ if it exists, x^* is unique and global (same proof)
3. f strongly convex $\Rightarrow x^*$ the global minimiser exists and is unique (coercive)

It is possible to show at least the first result above without assuming that f is differentiable, and taking only the definition of local minimum as

$$\exists \varepsilon > 0 \ \forall x \in B(x^*, \varepsilon) : f(x) \geq f(x^*).$$

Take $y \neq x$ arbitrary. Rearranging the convexity inequality to

$$f((1-t)x^* + ty) - f(x^*) \leq t(f(y) - f(x^*)) \quad (3)$$

and taking $t < \varepsilon/\|y - x^*\|$ gives

$$\|((1-t)x^* + ty) - x^*\| < t\|y - x^*\| < \varepsilon \Rightarrow (1-t)x^* + ty \in B(x^*, \varepsilon) \quad (4)$$

and hence from above and since x^* is a local minimum

$$0 \leq f(y) - f(x^*). \quad (5)$$

Therefore x^* is a global minimum.

Lemma 4. (First order characterisation) Let $f \in C^1(\mathbb{R}^p, \mathbb{R})$ be convex. Then

$$\forall x, y \in \mathbb{R}^p : f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle. \quad (6)$$

Moreover, if f is μ -strongly convex then

$$\forall x, y \in \mathbb{R}^p : f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|y - x\|^2. \quad (7)$$

For the convex case, the above lemma can be seen as a *supporting hyperplane* result if you consider that the *epigraph* of a convex function is a convex set. The epigraph consists of points in $\mathbb{R}^{p+1}$ and its members have the form $(f(x), x) \in \mathbb{R}^{p+1}$. The supporting hyperplane vector is then $(1, -\nabla f(x))$.

If $x^* \in \arg \min f$, then by Fermat's rule we have $\nabla f(x^*) = 0$. Using this in the equation above yields

$$\forall y \in \mathbb{R}^p : f(y) - f(x^*) \geq \frac{\mu}{2}\|y - x^*\|^2. \quad (8)$$

This is important because it means that any bound on the behaviour of $f(x) - f(x^*)$ automatically translates to a bound on $\|x - x^*\|^2$ – which is a critical connection for reasoning about convergence rates of algorithms.

Definition 5. (L -Lipschitz) A function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is L -Lipschitz if

$$\exists L > 0 \quad \forall x, y \in \mathbb{R}^p : |f(x) - f(y)| \leq L\|x - y\|. \quad (9)$$

Lipschitz functions are of course continuous, but it also turns out that they are differentiable *almost everywhere*. Where the derivative (or gradient) exist, the norm is bounded by L .

Definition 6. (L -smooth) A function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is L -smooth for $L > 0$ if ∇f is L -Lipschitz, that is

$$\forall x, y \in \mathbb{R}^p : |\nabla f(x) - \nabla f(y)| \leq L\|x - y\| \quad (10)$$

Lemma 7. (Quadratic Upper Bound) Let f be differentiable and L -smooth. Then

$$\forall x, y \in \mathbb{R}^p : \left| f(y) - f(x) - \langle \nabla f(x), y - x \rangle \right| \leq \frac{L}{2} \|y - x\|^2 \quad (11)$$

Proof. Idea: express $f(y) - f(x)$ as the integral of the gradient on the segment $[x, y]$ and then use the Lipschitzness to bound the gradient.

$$\begin{aligned} f(y) - f(x) &= \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle dt \quad (\text{FTC1 on } g(t) \text{ over } [0, 1]) \\ &= \left(\int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle dt \right) + \langle \nabla f(x), y - x \rangle \\ &\leq \left(\int_0^1 \| \nabla f(x + t(y - x)) - \nabla f(x) \| \|y - x\| dt \right) + \langle \nabla f(x), y - x \rangle \\ &= \left(\int_0^1 Lt \|y - x\|^2 dt \right) + \langle \nabla f(x), y - x \rangle \\ &= \frac{L}{2} \|y - x\|^2 + \langle \nabla f(x), y - x \rangle \end{aligned}$$

■

Corollary 8. If f is differentiable and L -smooth (Lemma 7), and additionally f is convex, then

$$0 \leq f(y) - f(x) - \langle \nabla f(x), y - x \rangle \quad (\text{convexity}) \quad (12)$$

and therefore from Lemma 7,

$$0 \leq f(y) - f(x) - \langle \nabla f(x), y - x \rangle \leq \frac{L}{2} \|y - x\|^2 \quad (13)$$

Setting $y = x - \frac{1}{L} \nabla f(x)$ (i.e. one gradient step) in the above Corollary, we have for all $x \in \mathbb{R}^p$:

$$f\left(x - \frac{1}{L} \nabla f(x)\right) - f(x) \leq -\frac{1}{2L} \|\nabla f(x)\|^2. \quad (14)$$

So one step of gradient descent is sure to decrease the function value by at least $\|\nabla f(x)\|^2 \times \text{a constant}$ (when f is convex).

Remark. (Strongly convex duality with L -smooth) Putting together the strongly convex inequality and the quadratic upperbound we have that for $f : \mathbb{R}^p \rightarrow \mathbb{R}$ strongly convex and L -smooth, if $x^* \arg \min f$ then

$$\forall x \in \mathbb{R}^p : \frac{\mu}{2} \|x - x^*\|^2 \leq f(x) - f(x^*) \leq \frac{L}{2} \|x - x^*\|^2. \quad (15)$$

So μ provides a lower bound on the function values and L proves an upper bound. Such functions f are therefore bounded by two quadratics.

Theorem 9. (Subgradient inequality under smoothness)

Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be such (13) holds (no another assumptions). Then we have the following inequality too:

$$\forall x, y \in \mathbb{R}^p : f(x) - f(y) \leq \langle \nabla f(x), x - y \rangle - \frac{1}{2L} \|\nabla f(x) - \nabla f(y)\|^2 \quad (16)$$

Proof. The key to the proof is to consider the point $z = x - \frac{1}{\beta}(\nabla f(x) - \nabla f(y))$. Writing $f(y) - f(x)$ as $f(y) - f(z) + f(z) - f(x)$ and then using the above inequality twice and substituting in the definition of z gives the result. ■

Corollary 10. (Inequality characterisation of convexity and L -smoothness)

Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$. Then f is convex and L -smooth if and only if

$$0 \leq f(y) - f(x) - \langle \nabla f(x), y - x \rangle \leq \frac{L}{2} \|y - x\|^2 \quad (17)$$

That is, (13) is a characterisation of convexity and L -smoothness.

Note that both inequalities here are important.

Proof. $\Rightarrow$ is [Corollary 8](#), and $\Leftarrow$ is [Theorem 9](#) (eval at (x, y) and (y, x) , then add the inequalities). ■

Gradient Descent Algorithm

Definition 11. (Gradient Descent) Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$. The gradient descent method generates the sequence $(x_k) \subseteq \mathbb{R}^p$ by:

$$\begin{aligned} x_0 &\in \mathbb{R}^p \text{ (arbitrary)} \\ x_{k+1} &= x_k - \alpha_k \nabla f(x_k) \quad \text{for } k = 0, 1, \dots \end{aligned} \quad (18)$$

where $(\alpha_k) > 0$ is called the ‘learning rate’ (can be dynamic or fixed).

The essential results on gradient descent under the typical assumptions on f are contained in the following theorem.

Theorem 12. (Gradient Descent for convex, L -smooth functions)

Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be a convex differentiable function that has a minimiser x^* and whose gradient is L -Lipschitz (f is L -smooth).

Gradient descent with a constant learning rate $\alpha_k = 1/L$ satisfies

convergence in value:

$$f(x_k) - f(x^*) \leq \frac{L\|x_0 - x^*\|^2}{2k} = O(1/k) \quad (19)$$

convergence in iterates:

$$x_k \rightarrow x_\infty \text{ as } k \rightarrow \infty \quad (20)$$

Proof. The proof of convergence in value is a more verbose and personal reproduction of that in [2], which is itself a simplified version of Nesterov's proof in his 2004 lecture notes.

From [Corollary 8](#) we know that a single gradient step satisfies

$$f\left(x - \frac{1}{L}\nabla f(x)\right) - f(x) \leq -\frac{1}{2L}\|\nabla f(x)\|^2. \quad (21)$$

or more generally

$$f(x_{k+1}) - f(x_k) \leq \frac{1}{2L}\|\nabla f(x_k)\|^2. \quad (22)$$

Define

$$\delta_k := f(x_k) - f(x^*) \text{ for } k = 0, 1, \dots \quad (23)$$

Then we have

$$\begin{aligned} \delta_{k+1} &\equiv f(x_{k+1}) - f(x^*) \stackrel{(22)}{\leq} f(x_k) - f(x^*) - \frac{1}{2L}\|\nabla f(x_k)\|^2 \\ &\equiv \delta_k - \frac{1}{2L}\|\nabla f(x_k)\|^2. \end{aligned} \quad (24)$$

Since f is convex, by the first order characterisation we have

$$\begin{aligned} \delta_k &\equiv f(x_k) - f(x^*) \leq \langle \nabla f(x_k), x_k - x^* \rangle \\ &\leq \|x_k - x^*\| \cdot \|\nabla f(x_k)\| \quad (\text{Cauchy Schwarz}) \end{aligned} \quad (25)$$

The next idea is to show that $\|x_k - x^*\|$ is weakly decreasing in k . It turns out that if we can show the above recurrence inequality will telescope to exactly what we want.

From [Corollay 10](#), we have (by same twice-eval-and-add trick noted in the proof):

$$\frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2 \leq \langle \nabla f(x) - \nabla f(y), x - y \rangle \quad (26)$$

Then, since $\nabla f(x^*) = 0$, we have

$$\begin{aligned} \|x_{k+1} - x^*\|^2 &= \left\| x_k - \frac{1}{L} \nabla f(x_k) - x^* \right\|^2 = \left\| (x_k - x^*) - \left(\frac{1}{L} \nabla f(x_k) \right) \right\|^2 \\ &= \|x_k - x^*\|^2 + \frac{1}{L^2} \|\nabla f(x_k)\|^2 - \frac{2}{L} \langle x_k - x^*, \nabla f(x_k) \rangle \\ &= \|x_k - x^*\|^2 + \frac{1}{L^2} \|\nabla f(x_k)\|^2 \\ &\quad - \frac{2}{L} \langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*) \rangle \\ &\leq \|x_k - x^*\|^2 + \frac{1}{L^2} \|\nabla f(x_k)\|^2 \\ &\quad - \frac{2}{L} \left(\frac{1}{L} \|\nabla f(x_k) - \nabla f(x^*)\|^2 \right) \quad (26) \\ &= \|x_k - x^*\|^2 + \frac{1}{L^2} \|\nabla f(x_k)\|^2 - \frac{2}{L^2} \|\nabla f(x_k)\|^2 \quad (\nabla f(x^*) = 0) \\ &= \|x_k - x^*\|^2 - \frac{1}{L^2} \|\nabla f(x_k)\|^2 \\ &\leq \|x_k - x^*\|^2 \end{aligned} \quad (27)$$

With $\|x_k - x^*\|$ proven to weakly decrease in k , the next idea is to use (24) and (25) together and find a telescoping sum. Squaring (25), we get

$$\frac{\delta_k^2}{\|x_k - x^*\|^2} \leq \|\nabla f(x_k)\|^2 \quad (28)$$

and since $\|x_k - x^*\| \leq \|x_0 - x^*\|$,

$$\frac{\delta_k^2}{\|x_0 - x^*\|^2} \leq \frac{\delta_k^2}{\|x_k - x^*\|^2} \leq \|\nabla f(x_k)\|^2. \quad (29)$$

Using this in (24) we get

$$\delta_{k+1} \leq \delta_k - \frac{\delta_k^2}{2L\|x_0 - x^*\|^2} = \delta_k - A\delta_k^2 \quad (30)$$

after letting $A := 1/(2L\|x_0 - x^*\|^2)$. The idea is now to reduce to a single index rather than depending on both k and $k + 1$.

$$A\delta_k^2 + \delta_{k+1} \leq \delta_k \quad (31)$$

Dividing by $1/\delta_k\delta_{k+1} > 0$,

$$A\frac{\delta_k}{\delta_{k+1}} + \frac{1}{\delta_k} \leq \frac{1}{\delta_{k+1}} \quad (32)$$

and then since $\delta_{k+1} \leq \delta_k$ (by 24),

$$A \leq \frac{1}{\delta_{k+1}} - \frac{1}{\delta_k} \quad (33)$$

and we have found our candidate for the telescoping sum. Summing over $k = 0, \dots, t - 1$,

$$tA \leq \sum_{k=0}^{t-1} \frac{1}{\delta_{k+1}} - \frac{1}{\delta_k} = \frac{1}{\delta_t} - \frac{1}{\delta_1} \leq \frac{1}{\delta_t}. \quad (34)$$

Unpacking the definition of A gives (19), and we have $f(x_k) \rightarrow f(x^*)$ at the $O(\frac{1}{k})$ rate.

Proving point coverage in this case is much easier. Let $r = \|x_0 - x^*\|$. We know a few things from the above work. First, the sequence $a_k := \|x_k - x^*\|$ converges to some $l \in [0, r]$. In particular, we do not yet know that $l = 0$.

Second, we know $x_k \in \overline{B}(x^*, r)$. Therefore (x_k) is a bounded sequence in $\mathbb{R}^p$, has at least one convergent subsequence by the Bolzano-Weierstrass theorem. Let this subsequence be $x_{\varphi(k)} \rightarrow x_s$ for $\varphi(k) \geq k$. By the continuity of f , we know $f(x_{\varphi(k)}) \rightarrow f(x_s)$. But the sequence

$$(f(x_{\varphi(k)}))_k \quad (35)$$

is itself a subsequence of the convergent sequence $f(x_k)$. This means that $f(x_s) = f(x^*)$, so $x_s \in \arg \min f$. Hence, we know $\|x_k - x_s\| \rightarrow l \in [0, r]$, since x_s is a minimiser. But $x_{\varphi(k)} \rightarrow x_s$, hence by *SLAP*, $x_k \rightarrow x_s \in \arg \min f$.

■

Bibliography

- [1] U. Jang and E. K. Ryu, “Point Convergence of Nesterov's Accelerated Gradient Method: An AI-Assisted Proof.” 2026.
- [2] S. Bubeck, “Convex Optimization: Algorithms and Complexity,” *Foundations and Trends® in Machine Learning*, vol. 8, no. 3–4, pp. 231–357, Nov. 2015, doi: [10.1561/22000000050](https://doi.org/10.1561/22000000050).